Worksheet 6-4 & 6-5 FTC Part 2 #1
AP Calculus

Name_____

Directions: Find $G'(x)$.

1. $G(x) = \int_0^x \sqrt{2t+1}dt$

2. $G(x) = \int_{-1}^{2x} \sin(t^2)\, dt$

3. $G(x) = \int_3^{x^2} -\frac{1}{t^3+2t}dt$

4. $G(x) = \int_0^{\sin(x)} (2t-1)^2 dt$

5. $G(x) = \int_2^{e^{2x}} (t^3+1)dt$

6. $G(x) = \int_{4x}^{7} \ln(t^2+4)\, dt$

Directions: Find the following in terms of x.

7. $\frac{d}{dx}\int_{-3}^{\ln(x)} \sin(t+1)dt$

8. $\frac{d}{dx}\int_1^{3x^2} \sqrt{1+t^2}dt$

9. $\frac{d}{dx}\int_{2x}^{1} (t^2-t)dt$

10. $\frac{d}{dx}\int_1^{\sqrt{x}} \frac{t^2}{t^4+1}\, dt$

11. Find $g'(x)$ given that $g(x) = \int_2^{\tan(x)} f(t)dt$ and $f(t) = \cos(t)$

12. Let $g(x) = \frac{d}{dx}\int_0^x \sqrt{t^2+9}\,dt$. What is $g(-4)$?

A.) -5 B.) -3 C.) 3 D.) 4 E.) 5

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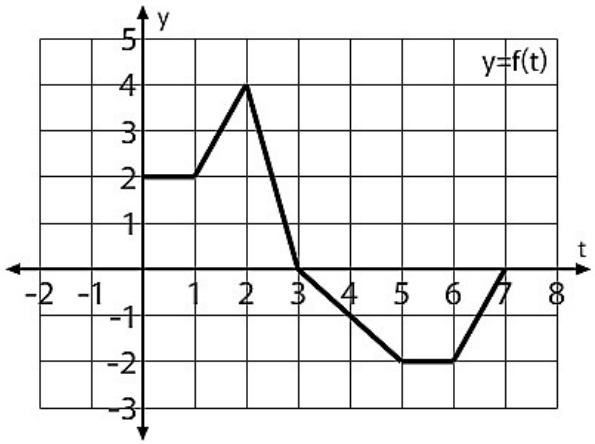
13. Let $g(x) = \int_0^x f(t)\,dt$, where f is the function whose graph is shown.

(a) Evaluate $g(0), g(1), g(2), g(3)$, and $g(6)$
(b) Evaluate $g'(2), g'(5)$, and $g'(7)$